

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 2

1. (Counting techniques)

- Motivation: Outcomes of equal probability, $\mathbb{P}(A) = |A|/|S|$.
- Rule of product: When an experiment consists of two parts, the first part of m distinct outcomes and the second part of n results, then we have total $m \times n$ outcomes
- Permutations: $P(n, k)$ how many ways of selecting k objects from n objects in an ordered manner.

$$P(n, k) = n \times (n - 1) \times \cdots (n - k + 1) = \frac{n!}{(n-k)!}$$

- Combinations: $C(n, k)$, how many ways of selecting k objects from n objects in an unordered manner

$$C(n, k) = P(n, k)/k!$$

- (a) (The Motivation) A four-digit PIN is selected randomly. What is the probability that there is no repeated digit?
- (b) (Difference between P and C) If we want to draw three names from a club of 18 members to be President, Vice-President, and Secretary, how many ways can this be done? What if we draw three names for three co-presidents?
- (c) You and your roommate pick 6 numbers between 1 and 48 and the 6 numbers cannot be repeated. Your roommate has to buy you dinner if at least 5 of the 6 numbers they picked match the numbers you picked and the order of numbers does not matter. What is the probability that you will get a free dinner?
- (d) What is the probability of getting exactly 2 Ace in 5 cards drawn randomly without replacement from a deck?

- **Solution:**

- (a) $\frac{P(10,4)}{10^4}$

- (b) $P(18, 3), C(18, 3)$

- (c) $\frac{C(6,5)C(42,1)+1}{C(48,6)}$

- (d) $\frac{C(4,2)C(48,3)}{C(52,5)}$

2. (Conditional Probability and Bayes' Theorem)

- Conditional Probability: $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$.
- Bayes' Rule: $\mathbb{P}(B|A) = \frac{\mathbb{P}(A|B) \cdot \mathbb{P}(B)}{\mathbb{P}(A)}$.
- Independent Events: $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$.
- You are given 1000 coins. Among them, one coin has head on both sides. The other 999 coins are fair coins. You randomly choose a coin and toss it 10 times. Each time, the coin turns up heads. What is the probability that the coin you choose is the unfair one?

- **Solution:**

- (a) Stage 1: choose a coin randomly; Stage 2: toss the coin 10 times

- (b) Define the event $A = \{\text{pick a unfair coin}\}$, $B = \{\text{all the 10 tosses are head}\}$.

(c) $\mathbb{P}(A) = 1/1000$, $\mathbb{P}(B|A) = 1$, $\mathbb{P}(B|A^c) = 1/1024$.

(d) Our target

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|A^c)\mathbb{P}(A^c)} = \frac{\frac{1}{1000}}{\frac{1}{1000} + \frac{\frac{999}{1000}}{1024}} = 0.506$$

3. (Discrete Random Variable)

- Probabilistic modeling: sample space and probability

You go to a casino and flip a fair coin.

(a) What is the sample space S ? Is it numerical? How likely are the outcomes?

- Random variable (r.v.): a mapping from the sample space S to real numbers $\mathbb{R}$.
You win \$1 for each head, and lose \$2 for each tail.

(b) What are your possible net gains? How likely are they?

(c) The net gain is a random variable X . Is it discrete or continuous?

(d) More examples:

- i. The amount of gasoline needed to drive your car 200 miles;
- ii. The number of kayaks in Lake Carnegie on any given day;

- Descriptions of a discrete random variable I: complete characterizations.

The Probability Mass Function (PMF) specifies the chance of each numerical result:

$$p(x) = \mathbb{P}(X = x) = \mathbb{P}(\{s \in S | X(s) = x\}) = \sum_{s: X(s)=x} \mathbb{P}(\{s\}), \quad x \in \mathbb{R}$$

(e) Find the function p for the coin-flipping example, and check the followings:

$$\begin{aligned} p(x) &\geq 0 & \forall x \in \Omega; \\ p(x) &= 0 & \text{otherwise,} \end{aligned}$$

where $\Omega = \{-2, 1\}$ is the range of the random variable X , and $\sum_{x \in \Omega} p(x) = 1$.

The Cumulative Distribution Function (CDF):

$$F(x) = \mathbb{P}(X \leq x) = \sum_{y: y \leq x} p(y)$$

(f) Find the function F in the coin-flipping example.

Draw its line graph, check its nonnegativity, boundedness and monotonicity.

- Descriptions of a discrete random variable II: summary statistics

The expectation quantifies the “average” value of a random variable:

$$\mathbb{E}X = \sum_{x \in \Omega} x \mathbb{P}(X = x) = \sum_{x \in \Omega} x p(x).$$

(g) What is your net gain on average? How to interpret that?

(h) Expectation of a function: $\mathbb{E}g(x) = \sum_{x \in \Omega} g(x)p(x)$

The variance quantifies the uncertainty of a random variable:

$$\text{var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}(X^2) - (\mathbb{E}X)^2$$

(i) The variance is the expectation of $g(x) = (x - \mathbb{E}X)^2$

x	0	1	2	3	4
$p(x)$	0.65	0.20	0.10	0.04	0.01

- (j) What is the variance of your net gain?
- (k) Another discrete example: Let X denote the number of broken eggs in a randomly selected dozen of eggs, with the distribution described below. Given a randomly selected dozen, find:
- probabilities of the following events: at most two broken eggs; fewer than two broken eggs; exactly ten unbroken eggs;
 - the expectation and the variance of X .

• **Solution:**

- (a) $S = \{H, T\}$, no, 0.5-0.5.
- (b) $\mathbb{P}(X = 1) = 0.5$, $\mathbb{P}(X = -2) = 0.5$
- (c) discrete.
- (d) discrete, continuous.
- (e) $p(1) = 0.5$, $p(-2) = 0.5$, their sum is 1.
- (f) $F(x) = 0, \text{ if } x < -2; F(x) = 0.5, \text{ if } -2 \leq x < 1; F(x) = 1, \text{ if } x \geq 1$. $F(x)$ is non-negative, bounded and increases monotonically (not strict).
- (g) -0.5; for large n , you lose roughly $\$n/2$ after n rounds.
- (j) $9/4$
- (k) i. 0.95, 0.85, 0.10; ii. 0.56, 0.8064.